

Ben Morris

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Career Objective

To combine my background in software engineering, robotics, and experience design to bridge digital systems and physical hardware - creating imaginative, technology-driven experiences in themed entertainment.

Education

Carnegie Mellon University

Aug 2025 - May 2027

Master of Entertainment Technology

Olin College of Engineering

Aug 2020 - May 2024

B.S. in Engineering with a Concentration in Computing | 3.80 GPA

Skills

Languages: Python, Java, C#, C, C++, Arduino, MATLAB, OCaml, Haskell, Swift, Javascript, SQL, SystemVerilog

Tools: Git/GitHub, Perforce, ROS2, Unity, Ubuntu, Computer Vision, LiDAR, ARKit, Azure ML, Flask, Django

Experience

Tutor Intelligence

Dec 2024 - Jul 2025

Robotics Software Engineer

- Designed and implemented new features to increase robustness in automatic palletizing applications
- Spearheaded effort to reduce mechanical failures in sensors, implementing automatic issue resolution and early warning signs for failure, causing mechanical sensor failures to drop more than 80%
- Revamped manipulation technique, increasing our grip by an average of 1.4x across products

RoboLab

May 2022 - Dec 2022

Hummingbird Software Subteam Lead

- Used ArduPilot in both Python and C to control a drone with Marvelmind localization
- Researched and implemented both localization and efficient path planning algorithms
- Created a custom real-time path planning algorithm for a swarm of drones which avoided collisions in a tight environment, and kept top speed 70% of the time with a margin of error of 3 cm

OCCAM Lab

May 2021 - Dec 2021

Software Researcher

- Integrated LiDAR, April Tags, & V/IO in Swift/Python to solve SLAM in an AR app for blind accessibility
- Built a real-time raycasting system to find navigational shortcuts within the generated map
- Moved project from proof-of-concept to functional prototype with 75% increase in accuracy

Relevant Projects

Franklin W. Olin Players: Served as president and acted, stage managed, and directed across six productions

Neato Tag: Group project to make vacuum robots play tag using LiDAR & computer vision in ROS2

Altitude: An immersive game in Unity meant to simulate the feel of skydiving with a full-body controller